

# oauth2-passkey

Passwordless Authentication Library for Rust

**Kimitoshi Takahashi (@ktaka)** | Tokyo Rust Show & Tell | 2026/03/31

`crates.io/crates/oauth2-passkey`    `crates.io/crates/oauth2-passkey-axum`

**Live Demo**

# Demo

- **Passkey Registration** - Create account with fingerprint/face
- **Passkey Login** - Authenticate without password
- **Google OAuth2 Login** - Sign in with Google
- **Account Linking** - Connect OAuth2 + Passkey to same user

[passkey-demo.ccmp.jp](https://passkey-demo.ccmp.jp)



# What the Demo Showed

| Feature                | Details                                                                        |
|------------------------|--------------------------------------------------------------------------------|
| <b>OAuth2/OIDC</b>     | Google login (also works with FedCM)                                           |
| <b>Passkey</b>         | Google Password Manager, Apple, Windows Hello, bitwarden, Proton Pass, YubiKey |
| <b>Account Linking</b> | OAuth2 + Passkey mapped to same user                                           |
| <b>Session</b>         | Cookie-based session with CSRF protection                                      |

All handled by a single library: `oauth2-passkey`

# Motivation

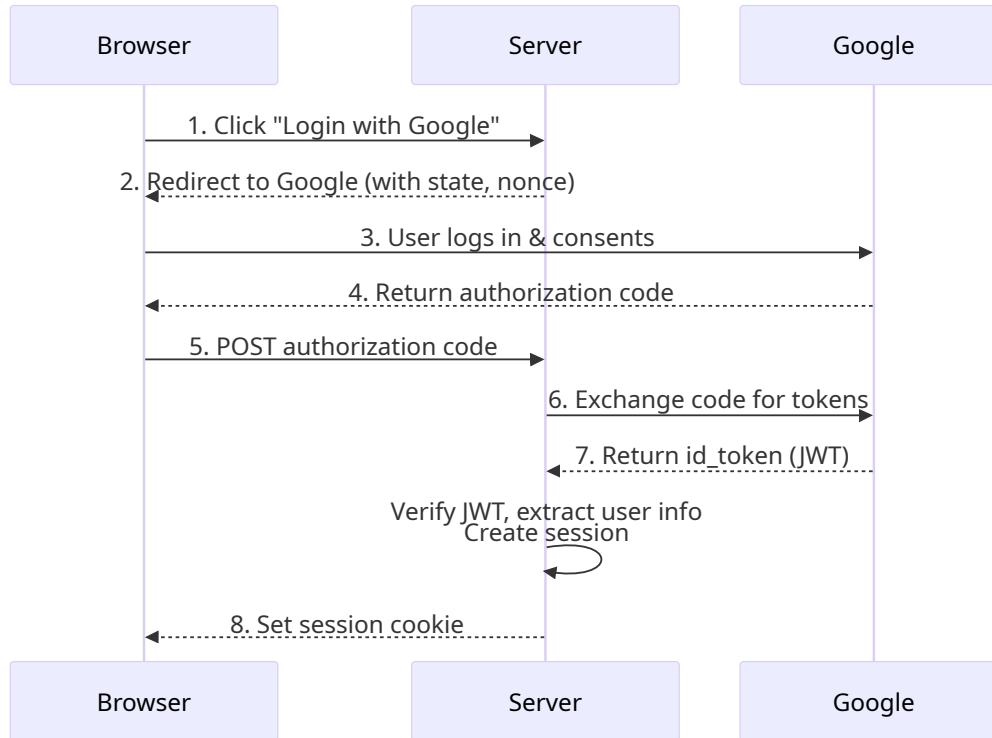
I wanted to build **exactly what you just saw** - and do it myself in Rust.

- Existing Rust ecosystem:
  - `webauthn-rs` - WebAuthn only
  - Various OAuth2 crates - OAuth2 only
  - **No combined solution** with session management
- So I built one. Published on **crates.io**.

# Agenda

1. **How OAuth2 & Passkey Work** - What happened behind the demo
2. **Using the Library** - init, extractor, middleware
3. **Storage & LazyLock** - Multi-DB support, why not Axum State
4. **Integrating with Your App** - Linking your data to auth users
5. **Wrap-up** - Summary & links

# How OAuth2/OIDC Works

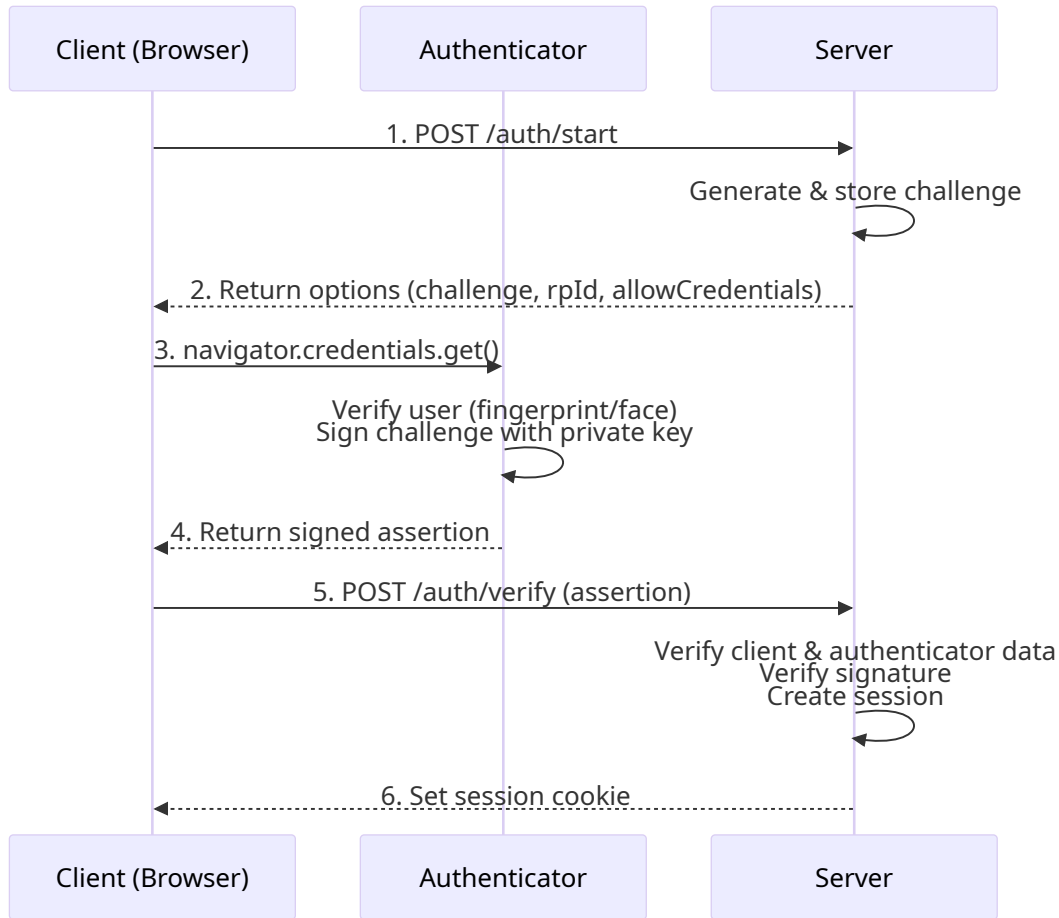


## Page-redirect based auth:

1. User clicks "Login with Google"
2. Redirect to Google consent screen
3. Google returns authorization code
4. Server exchanges code for **id\_token** (JWT)
5. Extract user info, create session
6. Set session cookie

# How Passkey/WebAuthn Works





## JavaScript-driven, no redirects:

1. Server generates **challenge**
2. Browser calls `navigator.credentials.get()`
3. Authenticator signs challenge with **private key**
4. Server verifies with stored **public key**
5. Create session, set cookie

*Authenticators: Google  
Password Manager, YubiKey,  
Touch ID, Windows Hello*

# Using the Library

# Setup: init + merge

```
use oauth2_passkey_axum::{
    AuthUser, oauth2_passkey_full_router,
};

#[tokio::main]
async fn main() -> Result<(), Box<dyn std::error::Error>> {
    dotenv().ok();
    oauth2_passkey_axum::init().await?;           // 1. Initialize

    let app = Router::new()
        .route("/", get(index))
        .merge(oauth2_passkey_full_router()); // 2. Merge router

    // Auth endpoints are now at /o2p/*
    spawn_http_server(3001, app).await?;
    Ok(())
}
```

# Page Protection: AuthUser Extractor

```
// Required auth - redirects to login if not authenticated
async fn protected(user: AuthUser) -> impl IntoResponse {
    format!("Hello, {}!", user.label)
}

// Optional auth - allows anonymous access
async fn public(user: Option<AuthUser>) -> impl IntoResponse {
    match user {
        Some(u) => format!("Hello, {}!", u.label),
        None    => "Hello, anonymous!".to_string(),
    }
}
```

Implemented via Axum's `FromRequestParts` trait:

- `AuthUser` -> auto-redirect on failure
- `Option<AuthUser>` -> no redirect, `None` for anonymous

# Page Protection: Middleware

```
let app = Router::new()
  // Protect single route
  .route("/secret", get(handler)
    .route_layer(from_fn(
      is_authenticated_redirect)))
  // Protect entire group
  .nest("/api", api_router()
    .route_layer(from_fn(
      is_authenticated_user_redirect))));
```

| Middleware     | Unauth | User? |
|----------------|--------|-------|
| _401           | 401    | No    |
| _redirect      | Login  | No    |
| _user_401      | 401    | Yes   |
| _user_redirect | Login  | Yes   |

All prefixed with `is_authenticated`

# **Storage & LazyLock Pattern**

# Multi-DB Support with sqlx

## DataStore trait:

```
pub(crate) trait DataStore: Send + Sync {  
    fn as_sqlite(&self)  
        -> Option<&Pool<Sqlite>>;  
    fn as_postgres(&self)  
        -> Option<&Pool<Postgres>>;  
    fn as_mysql(&self)  
        -> Option<&Pool<MySQL>>;  
}
```

One trait, three implementations. No runtime downcasting needed.

## Dispatch pattern:

```
let store = GENERIC_DATA_STORE  
    .lock().await;  
  
match (store.as_sqlite(),  
        store.as_postgres(),  
        store.as_mysql()) {  
    (Some(pool), _, _) =>  
        do_sqlite(pool).await?,  
    (_, Some(pool), _) =>  
        do_postgres(pool).await?,  
    (_, _, Some(pool)) =>  
        do_mysql(pool).await?,  
    _ => return Err(...),  
}
```



# Why LazyLock Instead of Axum State?

## Typical Axum State pattern:

```
struct AppState { db: PgPool, cache:
Redis }
let app =
Router::new().with_state(state);

// EVERY handler needs State
async fn handler(
    State(s): State<AppState>
) { ... }

// 80+ internal functions need &AppState
async fn internal_fn(
    state: &AppState
) { ... }
```

## oauth2-passkey: LazyLock globals

```
static GENERIC_DATA_STORE:
    LazyLock<Mutex<Box<dyn DataStore>>>
    = LazyLock::new(|| {
        match store_type {
            "sqlite"    => Box::new(...),
            "postgres"  => Box::new(...),
            "mysql"     => Box::new(...),
        }
    });

// Any function can access directly
let store = GENERIC_DATA_STORE
    .lock().await;
```

# LazyLock: Benefits & Trade-offs

## Benefits

- **Zero boilerplate for users** - NO AppState struct to create
- **No state threading** - internal functions access storage directly
- **Env-only config** - Set `GENERIC_DATA_STORE_TYPE=postgres` in `.env`
- **Fail-fast init** - `init().await?` forces evaluation at startup

## Trade-offs

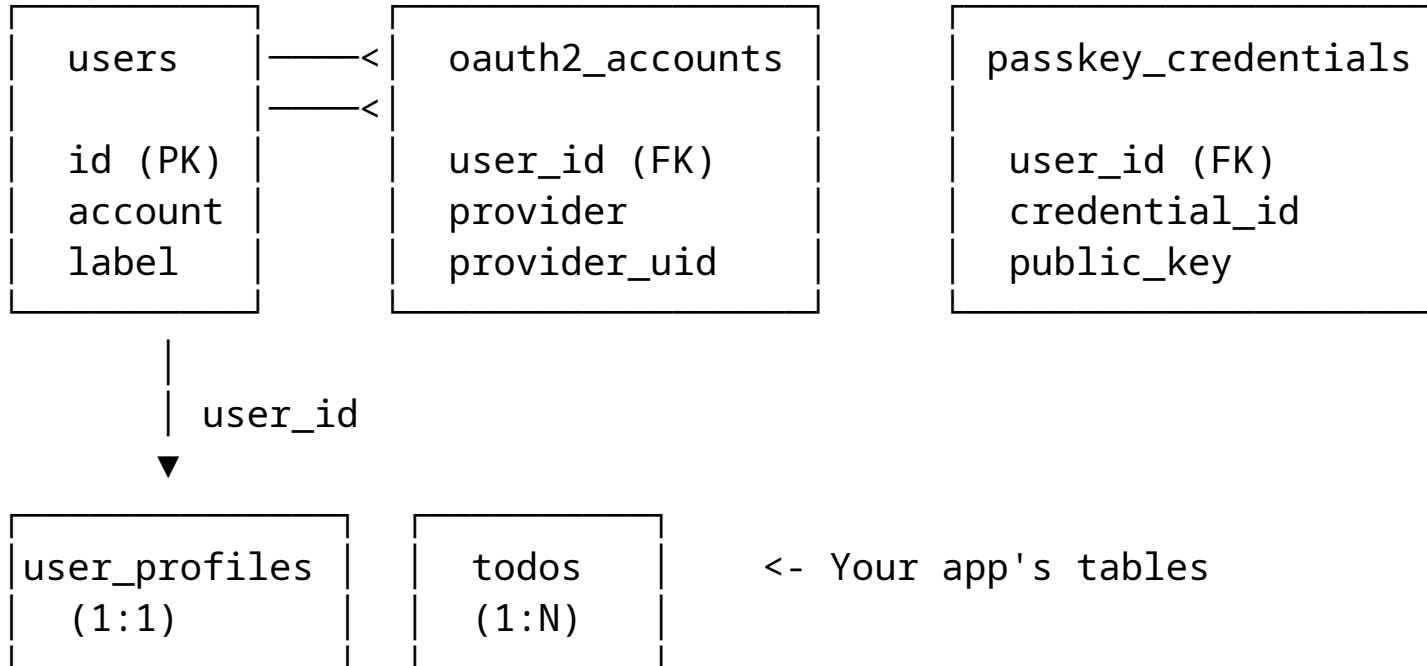
- Single instance per process (fine for auth library)
- Implicit dependencies (function signatures don't show DB access)
- Test isolation needs `#[serial]` (shared global state)

A library that requires users to manage `AppState` is harder to adopt. LazyLock keeps complexity **inside** the library.

# **Integrating with Your App**

# Account Linking: Internal DB Structure

oauth2-passkey manages these tables:



One user can have multiple OAuth2 accounts AND multiple passkeys.

# Your App's Data: Link via AuthUser.id

## Setup: two databases, one app

```
// 1. oauth2-passkey init
oauth2_passkey_axum::init().await?;

// 2. App's own DB
let pool = db::init_db().await?;
let state = AppState { pool };

// 3. Combine routes
let app = Router::new()
    .route("/", get(index))
    .merge(handlers::router())
    .with_state(state)
    .merge(oauth2_passkey_full_router());
```

## Handler: use user.id as FK

```
async fn create_todo(
    State(state): State<AppState>,
    user: AuthUser,
    Form(form): Form<TodoForm>,
) -> Result<Response, ...> {
    db::create_todo(
        &state.pool,
        &user.id, // link to auth user
        &form.title,
    ).await?;
    Ok(Redirect::to("/").into_response())
}
```

See demo-profile (1:1) and demo-todo  
(1:N).

**Wrap-up**



# Summary

| What         | Details                                    |
|--------------|--------------------------------------------|
| Library      | oauth2-passkey + oauth2-passkey-axum       |
| What it does | Passkey + OAuth2 auth for Axum apps        |
| DB support   | SQLite, PostgreSQL, MySQL (via sqlx)       |
| Key design   | LazyLock globals, DataStore trait dispatch |
| Integration  | AuthUser.id links your app data to auth    |



GitHub

## Links

- **crates.io:** [oauth2-passkey](#), [oauth2-passkey-axum](#)
- **GitHub:** [github.com/ktaka-ccmp/oauth2-passkey](https://github.com/ktaka-ccmp/oauth2-passkey)
- **X:** @ktaka

# Thank You!

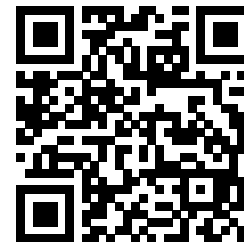
Questions?

**Kimitoshi Takahashi (@ktaka)** on X / GitHub

# About Me

- **@ktaka** (X / GitHub)
- Self-employed, reskilling in Rust
- Building web authentication libraries
- Third year writing Rust

[ktaka.blog.ccmp.jp/p/p.html](https://ktaka.blog.ccmp.jp/p/p.html)



Contact